MH4110 Partial Differential Equations

Chapter 6 - Harmonic Functions

Outline

This chapter is devoted to the Laplace's equation. We shall mainly focus on one of its important properties, i.e., the so-called *maximum principle*, and solve the equation in rectangular and circular domains by separation of variables.

- Laplace's equation & Maximum principle
- Separation of variables in rectangles and cubes
- Poisson's formula

§6.1 Laplace's equation

Laplace equation

If a diffusion or wave process is stationary (independent of time), i.e., $u_t = 0$ and $u_{tt} = 0$, then both the diffusion and the wave equations reduce to the *Laplace's equation*:

$$\Delta u = 0, \tag{1}$$

where in one, two and three dimensions:

$$u_{xx} = 0, \quad u_{xx} + u_{yy} = 0, \quad u_{xx} + u_{yy} + u_{zz} = 0. \tag{2}$$

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Note that we used the conventional notation (in 3D):

$$\begin{aligned}
&\nabla u = (\partial_x, \partial_y, \partial_z)u = (u_x, u_y, u_z),\\
&\mathrm{div}\mathbf{u} = \nabla \cdot \mathbf{u} = (\partial_x, \partial_y, \partial_z) \cdot (u, v, w) = u_x + v_y + w_z,\\
&\Delta u = \nabla \cdot (\nabla u) = u_{xx} + u_{yy} + u_{zz}.
\end{aligned}$$

§6.1 Laplace's equation

Definition: Harmonic function

A solution of the Laplace's equation

$$\Delta u = 0$$

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- In one dimension, the linear function $u(x) = ax + b$ is the only harmonic function. $u_{xx} = 0 \Rightarrow u_x = a \Rightarrow u(x) = ax + b$
- We know from complex analysis that an analytic function of complex variable

$$f(z) = u(z) + \mathrm{i}v(z) = u(x + \mathrm{i}y) + \mathrm{i}v(x + \mathrm{i}y),$$

where u, v are real-valued functions, satisfies the Cauchy-Riemann equations:

$$u_x = v_y, \quad u_y = -v_x.$$

Differentiating them we find that $\Delta u = 0, \quad \Delta v = 0.$

Thus, the real and imaginary parts of an analytic function are harmonic.

§6.1 Laplace's equation

Definition: Poisson's equation

The inhomogeneous version of (1):

$$\Delta u = f,$$

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Boundary value problem

Let D be a given domain with boundary denoted by $\mathrm{bdy}(D)$. The Laplace's equation with boundary conditions generally takes the form:

$$\begin{aligned} &\Delta u = f, \quad \text{in } D, \\ &u = h \ \text{ or } \ \frac{\partial u}{\partial \mathbf{n}} = h \ \text{ or } \ \frac{\partial u}{\partial \mathbf{n}} + cu = h \ \text{ on } \ \mathrm{bdy}(D), \end{aligned} \tag{3}$$

D-type

where $\mathbf{n}$ is the unit outer normal along $\mathrm{bdy}(D)$.

Theorem: Maximum principle

Let D be a (simply or multiply) connected bounded open set (in either two- or three- dimensional space). Let either $u(x, y)$ or $u(x, y, z)$ be a harmonic function in D that is continuous on $\bar{D} = D \cup \mathrm{bdy}(D)$. Then the maximum and the minimum values of u are attained on $\mathrm{bdy}(D)$ and nowhere inside (unless u is a constant). In other words,

$$\max_{\mathbf{x}\in\bar{D}} u(\mathbf{x}) = \mathbf{u}(\mathbf{x_M}), \quad \min_{\mathbf{x}\in\bar{\mathbf{D}}} \mathbf{u}(\mathbf{x}) = \mathbf{u}(\mathbf{x_m}),$$

for some $\mathbf{x}_M$ and $\mathbf{x}_m$ on the boundary $\mathrm{bdy}(D)$. Moreover, they can not be attained inside D unless u is a constant.

$$B = \{ \vec{x} \in R^3 \mid |\vec{x}| < a \}$$

$$\partial B = \{ \vec{x} \in R^3 \mid |\vec{x}| = a \}$$

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- Consider the one-dimensional harmonic functions: $u(x) = ax + b$ on any finite interval.

A Simply connected bounded open set is (c, d)

§6.1 Laplace's equation

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- Consider the one-dimensional harmonic functions: $u(x) = ax + b$ on any finite interval.
- We can use Maximum principle to show the uniqueness of solutions to Laplace/Poisson's equations.

Uniqueness of solutions to Laplace/Poisson equations

To prove the uniqueness, suppose that

$$\begin{aligned} \Delta u &= f, \quad \text{in } D \qquad & \Delta v &= f, \quad \text{in } D \\ u &= h \text{ on bdy } D \qquad & v &= h \text{ on bdy } D. \end{aligned} \tag{4}$$

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We want to show that $u \equiv v$ in D. So we simply subtract equations and let $w = u - v$. Then $\Delta w = 0$ in D and $w \equiv 0$ on bdy D. By the maximum principle

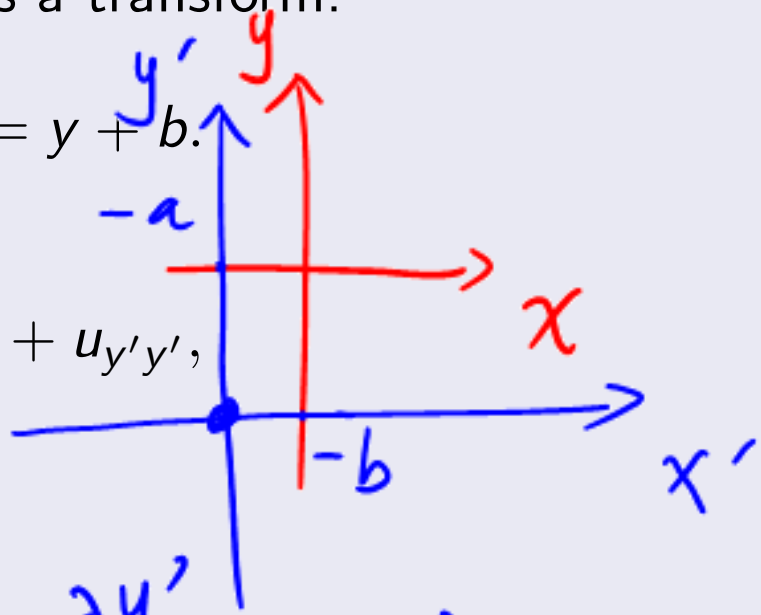
$$0 = w(\mathbf{x}_m) \le w(\mathbf{x}) \le w(\mathbf{x}_M) = 0 \quad \text{for all } \mathbf{x} \in D.$$

Therefore, both the maximum and minimum of $w(\mathbf{x})$ are zero. This means that $w \equiv 0$ and $u \equiv v$.

Invariance of Laplace operator under rigid motion

The Laplace operator is invariant under the translation and rotation. We look at 2D case. A translation (shifting) is a transform:

$$x' = x + a, \quad y' = y + b.$$



Invariance means:

$$u_{xx} + u_{yy} = u_{x'x'} + u_{y'y'},$$

which can be verified directly.

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial x'}\frac{\partial x'}{\partial x} + \frac{\partial}{\partial y'}\frac{\partial y'}{\partial x} = \frac{\partial}{\partial x'},$$

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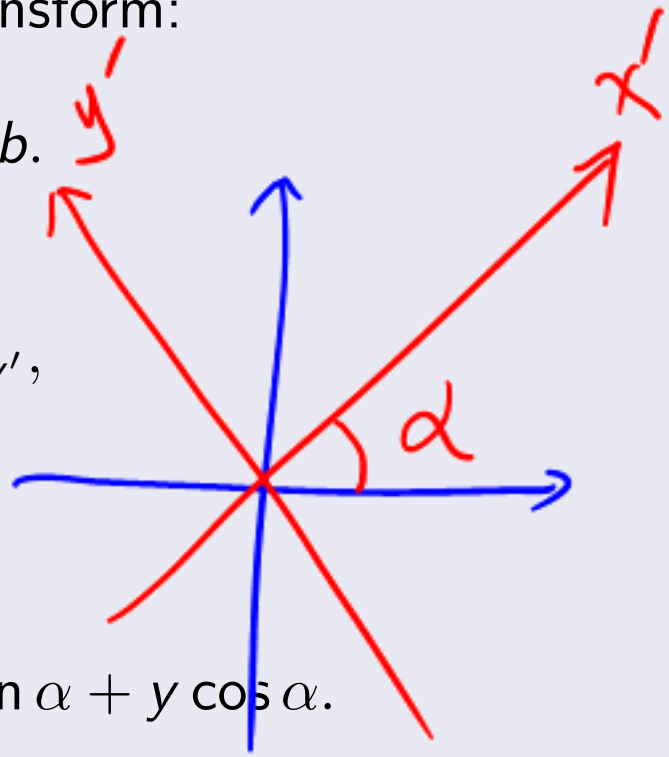

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A rotation of angle α is

$$x' = x\cos\alpha + y\sin\alpha, \quad y' = -x\sin\alpha + y\cos\alpha.$$

We can also verify that under this transform, the Laplace operator is invariant.

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial x'}\frac{\partial x'}{\partial x} + \frac{\partial}{\partial y'}\frac{\partial y'}{\partial x} = \cos\alpha\frac{\partial}{\partial x'} - \sin\alpha\frac{\partial}{\partial y'}$$

§6.1 Laplace's equation

Under the polar coordinates:

$$x = r\cos\theta, \quad y = r\sin\theta.$$

we can find that

$$\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = \frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial \theta^2}. \tag{5}$$

$$r = \sqrt{x^2+y^2} \qquad \theta = \operatorname{atan}\frac{y}{x}$$

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial r}\cdot\frac{\partial r}{\partial x} + \frac{\partial}{\partial \theta}\frac{\partial \theta}{\partial x}$$

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We consider special harmonic functions $u(r,\theta)$, independent of θ, i.e., $u(r,\theta) = u(r)$, so by (5),

$$\Delta u = u_{rr} + \frac{1}{r}u_r,$$

which can be solved

$$u_{rr} + \frac{1}{r}u_r = 0 \Rightarrow$$

$$ru_r = c_1$$

$$u_r = \frac{c_1}{r}$$

$$\Leftarrow (ru_r)_r = 0 \quad \Rightarrow \quad u = c_1 \log r + c_2.$$

$$r u_{rr} + u_r = 0$$

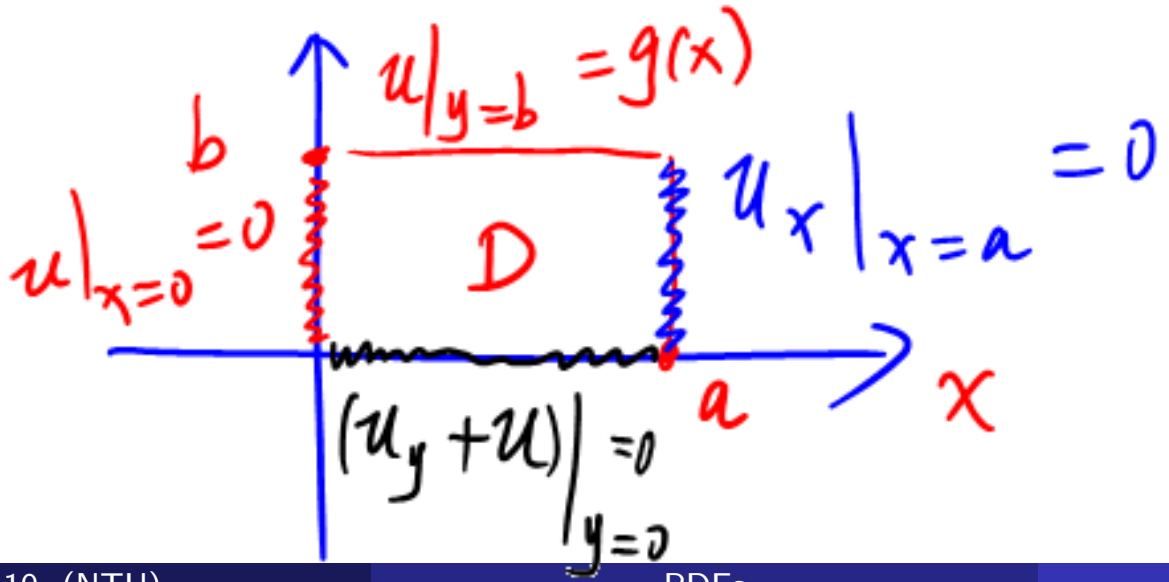
§6.2 Rectangles and cubes

Special geometries can be solved by separating the variables.

Laplace equation in a rectangle

Use separation of variables to solve the problem

$$
\begin{aligned}
&\Delta u = u_{xx} + u_{yy} = 0, \quad (x, y) \in D := (0, a) \times (0, b), \\
&u|_{x=0} = 0, \quad u_x|_{x=a} = 0 \quad \text{(left and right boundaries)}, \\
&(u_y + u)|_{y=0} = 0, \quad u|_{y=b} = g(x) \quad \text{(bottom and top boundaries)}.
\end{aligned}
\tag{6}
$$



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$X(x)Y'(0) + X(x)Y(0) = 0 \implies Y'(0) + Y(0) = 0$

As before, we look for a **separated solution** of (6) of the form

$$u(x, y) = X(x)Y(y), \tag{7}$$

which should satisfy the boundary conditions. This form will be substituted into (6) and result in ODEs for X and Y, respectively.

§6.2 Rectangles and cubes

Step I: Plugging (7) into (6) leads to

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Here λ must be a constant, and $X(x)$ is the solution to the eigen-problem:

$$X''(x) + \lambda X(x) = 0, \quad X(0) = X'(a) = 0. \tag{10}$$

$$\begin{cases} \lambda < 0 & X(x) = 0 \quad \text{trivial} \\ \lambda = 0 & X''(x) = 0 \quad X(x) = cx + d \\ & X(0) = X'(a) = 0 \Rightarrow c = d = 0 \end{cases}$$

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For a nontrivial solution, the boundary conditions $X(0) = X'(a) = 0$ force λ to be positive. Therefore, The solutions are

$$\lambda_n = \left(n + \frac{1}{2}\right)^2 \frac{\pi^2}{a^2}, \quad n = 0, 1, 2, \cdots, \tag{11}$$

$$X_n(x) = \sin \beta_n x, \quad \beta_n = \sqrt{\lambda_n}. \tag{12}$$

§6.2 Rectangles and cubes

Step II: we look at the y variable:

$$Y''(y) - \lambda Y(y) = 0, \quad Y'(0) + Y(0) = 0, \tag{13}$$

where we will impose the inhomogeneous boundary conditions at $y = b$ in the last step.

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where we will impose the inhomogeneous boundary conditions at $y = b$ in the last step.

Then we can get the general solution: $\beta_n = \sqrt{\lambda_n}$

$$Y_n(y) = c_1 e^{\beta_n y} + c_2 e^{-\beta_n y} = A_n \cosh(\beta_n y) + B_n \sinh(\beta_n y). \tag{14}$$

Note that it is more convenient to write the solution in cosh and sinh form

$$\sinh y = \frac{e^y - e^{-y}}{2}, \quad \cosh y = \frac{e^y + e^{-y}}{2}.$$

$$A_n \frac{e^{\beta_n y} + e^{-\beta_n y}}{2} + B_n \frac{e^{\beta_n y} - e^{-\beta_n y}}{2} = Y_n(y)$$

$$c_1 = (A_n + B_n)/2 \quad c_2 = (A_n - B_n)/2$$

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$Y_n'(0) = B_n \beta_n$

$Y_n(0) = A_n$

Then from the boundary condition at $y = 0$, we find $B_n \beta_n + A_n = 0$. Then we obtain

$$Y_n(y) = B_n\big(- \beta_n \cosh(\beta_n y) + \sinh(\beta_n y)\big). \tag{15}$$

Step III: We now assemble the solution

$Y_n(y)$

$$u(x,y) = \sum_{n=0}^{\infty} B_n \sin(\beta_n x)\big(-\beta_n \cosh(\beta_n y) + \sinh(\beta_n y)\big). \tag{16}$$

$X_n(x)$

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$$u(x,y) = \sum_{n=0}^{\infty} B_n \sin(\beta_n x)\big(-\beta_n \cosh(\beta_n y) + \sinh(\beta_n y)\big). \tag{16}$$

It remains to determine B_n by using the boundary condition: $u(x,b) = g(x)$. This requires that

$$g(x) = \sum_{n=0}^{\infty} \underline{\underline{B_n\big(-\beta_n \cosh(\beta_n b) + \sinh(\beta_n b)\big)}} \sin(\beta_n x),$$

Fourier sine series

By Chapter 5, we find

$$B_n = \frac{2}{a\big(-\beta_n \cosh(\beta_n b) + \sinh(\beta_n b)\big)} \int_0^a g(x)\sin(\beta_n x)dx. \tag{17}$$

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In summary, the unique solution of (6) is given by (16)-(17).

§6.3 Poisson's formula

The rotational invariance of Δ provides a hint that the circle is a natural shape for harmonic functions.

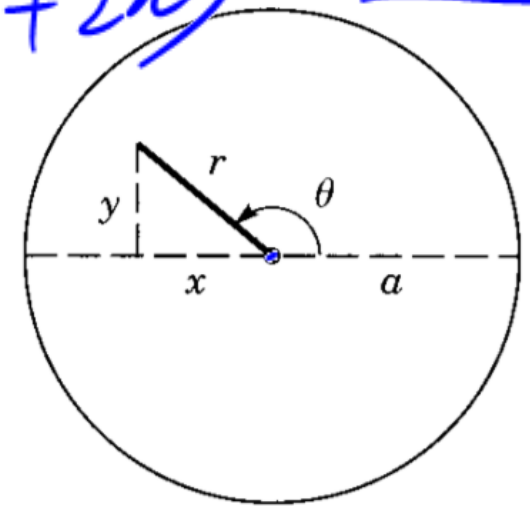
Example

Use separation of variables in polar coordinates to solve the Laplace's equation with Dirichlet boundary condition:

$$\Delta u = 0, \quad (x, y) \in D := \left\{(x, y) : r = \sqrt{x^2 + y^2} < a\right\},$$
$$u|_{r=a} = h(\theta), \quad \theta \in [0, 2\pi].$$

$\theta \in [0, 2\pi]$

$h(\theta) = h(\theta + 2\pi) \Longrightarrow h(0) = h(2\pi)$



§6.3 Poisson's formula

Solution: We first write the Laplace equation in polar coordinates $x = r\cos\theta$, $y = r\sin\theta$ as

$$\Delta u = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} = 0, \quad \underline{\underline{0 < r < a}}, \quad \theta \in [0, 2\pi]. \tag{18}$$

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We proceed with the normal process for separation of variables. Look for a separated (nontrivial) solution:

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Inserting it into the equation (18) leads to

$$R''(r)\Theta(\theta) + \frac{1}{r}R'(r)\Theta(\theta) + \frac{1}{r^2}R(r)\Theta''(\theta) = 0,$$

$$\left[-R''(r) - \frac{1}{r}R(r)\right]\Theta(\theta) = \frac{1}{r^2}R(r)\Theta''(\theta)$$

or

$$-\frac{r^2R''(r) + rR'(r)}{R(r)} = \frac{\Theta''(\theta)}{\Theta(\theta)} = -\lambda.$$

We can show that λ is a non-negative constant.

§6.3 Poisson's formula

Let us solve the problem in Θ first:

$$\Theta''(\theta) + \lambda\Theta(\theta) = 0, \quad \Theta(\theta) = \Theta(\theta + 2\pi),$$

where the periodic boundary condition is natural. key observation

$\lambda < 0$

$$\left.\begin{aligned} \Theta(\theta) &= c_1 e^{\sqrt{-\lambda}\,\theta} + c_2 e^{-\sqrt{-\lambda}\,\theta} \\ \Theta(\theta) &= \Theta(\theta + 2\pi) \end{aligned}\right\} \Rightarrow c_1 = c_2 = 0$$

$$\lambda = 0 \Rightarrow \left.\begin{aligned} \Theta(\theta) &= c_1\theta + c_2 \\ \Theta(\theta) &= \Theta(\theta + 2\pi) \end{aligned}\right\} \Rightarrow c_1 = 0$$

§6.3 Poisson's formula

$\lambda > 0 \qquad \Theta(\theta) = C_1 \cos\sqrt{\lambda}\,\theta + C_2 \sin\sqrt{\lambda}\,\theta$

Let us solve the problem in Θ first:

$C_1 \cos\sqrt{\lambda}\,\theta + C_2 \sin\sqrt{\lambda}\,\theta$

$$\Theta''(\theta) + \lambda\Theta(\theta) = 0, \quad \Theta(\theta) = \Theta(\theta + 2\pi),$$

$= C_1 \cos\sqrt{\lambda}(\theta + 2\pi) + C_2 \sin\sqrt{\lambda}(\theta + 2\pi)$

where the periodic boundary condition is natural.

$\Downarrow$

We find that the eigenvalues are

$$\lambda_n = n^2, \quad n = 0, 1, 2, \cdots,$$

$\sqrt{\lambda} = n$

and the corresponding eigenfunctions are

$$\Theta_n = \begin{cases} A_0, & \text{for } n = 0, \\ A_n \cos n\theta + B_n \sin n\theta, & \text{for } n \geq 1. \end{cases}$$

We can check if $\lambda < 0$ there is only one trivial solution $\Theta(\theta) = 0$.

§6.3 Poisson's formula

Now, we turn to the equation in $R(r)$.

- For $n = 0$,

$$r^2 R_0''(r) + rR_0'(r) = 0 \quad \Rightarrow \quad R_0(r) = C_0 + D_0 \log r.$$

$$\Downarrow$$

$$r R_0''(r) + R_0'(r) = 0$$

$$\Downarrow$$

$$\left(r R_0'(r) \right)' = 0$$

$$r R_0'(r) = D_0 \quad \Rightarrow R_0'(r) = \frac{D_0}{r}$$

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$$r^2 R_0''(r) + rR_0'(r) = 0 \quad \Rightarrow \quad R_0(r) = C_0 + D_0 \log r.$$

- For $n \geq 1$,

$$r^2 R''(r) + rR'(r) - n^2 R(r) = 0.$$

It is a Euler's equation with solutions of the form $R(r) = r^\alpha$ (α needs to be determined). Sine $\lambda = n^2$, the equation for R reduces to

$$\alpha(\alpha - 1)r^\alpha + \alpha r^\alpha - n^2 r^\alpha = 0,$$

$\alpha^2 r^\alpha - n^2 r^\alpha = 0 \Rightarrow \alpha^2 = n^2$

giving that $\alpha = \pm n$. Thus, the general solution is

$$R_n(r) = C_n r^n + D_n r^{-n}.$$

§6.3 Poisson's formula

We are ready to assemble the solution:

$$u(r,\theta) = \underbrace{C_0 + D_0 \log r}_{R_0(r)} + \sum_{n=1}^{\infty} \underbrace{[A_n \cos(n\theta) + B_n \sin(n\theta)]}_{\Theta_n(\theta)} \underbrace{\left[C_n r^n + D_n r^{-n}\right]}_{R_n(r)}.$$

§6.3 Poisson's formula

We are ready to assemble the solution:

$$u(r,\theta) = C_0 + D_0 \log r + \sum_{n=1}^{\infty} \left[A_n \cos(n\theta) + B_n \sin(n\theta)\right] \left[C_n r^n + D_n r^{-n}\right].$$

Physically, we would like to have the solution satisfying: $u(0,\theta) = \text{finite}$. The requirement that the solutions are finite is the boundary condition at $r = 0$. This implies

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The arbitrary constant C_n $(n > 0)$ can be absorbed by A_n and B_n without losing any information. Thus, we have

$$u(r,\theta) = C_0 + \sum_{n=1}^{\infty} r^n \big(A_n \cos(n\theta) + B_n \sin(n\theta)\big). \tag{19}$$

§6.3 Poisson's formula

Finally, we impose the boundary condition: $u(a, \theta) = h(\theta)$

$$h(\theta) = C_0 + \sum_{n=1}^{\infty} a^n \big(A_n \cos(n\theta) + B_n \sin(n\theta)\big).$$

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This is precisely the full Fourier series for $h(\theta)$, so we know that

$$A_n = \frac{1}{\pi a^n} \int_0^{2\pi} h(\phi) \cos(n\phi) d\phi \tag{20}$$

and

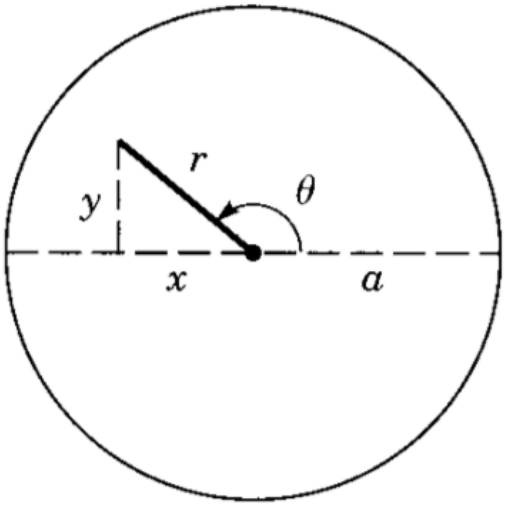
$$B_n = \frac{1}{\pi a^n} \int_0^{2\pi} h(\phi) \sin(n\phi) d\phi. \tag{21}$$

Equations (19) to (21) constitute the full solution of our problem.

§6.3 Poisson's formula

Now comes an amazing fact. The series (19) can be summed explicitly!

$$u(r,\theta) = (a^2 - r^2)\int_0^{2\pi} \frac{h(\phi)}{a^2 - 2ar\cos(\theta-\phi) + r^2}\frac{d\phi}{2\pi}. \tag{22}$$



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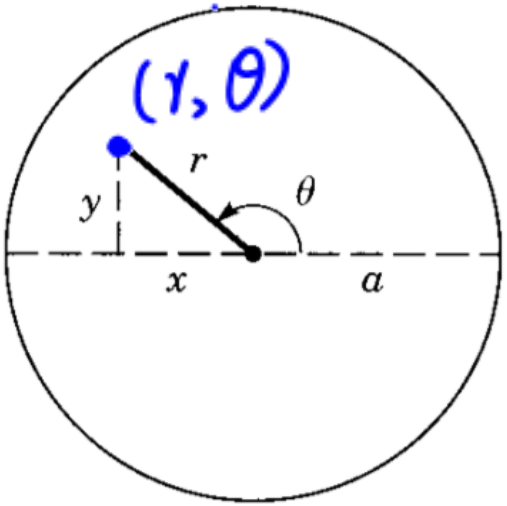

a weighted average of the boundary values

This single formula (22), known as Poisson's formula, replaces the triple of formulas (19) to (21). It expresses any harmonic function inside a circle in terms of its boundary values.